WORKSHEET · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 10: Recursion: Blocks That Call Themselves

Module 4 | 45 minutes | Recursion | Grades 6–8 · ages 12–14

Name: Class: Date:

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. recursion ___ A. A shape made of smaller copies of itself.
b. base case ___ B. The step that calls itself with a smaller input.
c. recursive case ___ C. The simple case where recursion stops.
d. fractal ___ D. A procedure that calls itself.

2. Plan your code

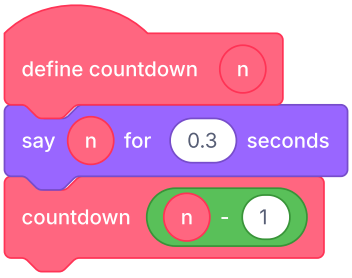
Before you build it: Recursive countdown: define countdown (n). If n > 0, say n for 0.3 seconds and call countdown (n − 1); otherwise say “Liftoff!”. Call countdown (5) on the flag. Write or draw the blocks you will use.

3. Plan your code

Before you build it: Fractal tree: define branch (size). If size > 10: move size, turn left 25, branch (size × 0.65), turn right 50, branch (size × 0.65), turn left 25, then move back −size. Point up, pen down and call branch (60). Write or draw the blocks you will use.

4. Debug the recursion

This recursive block runs until Scratch stops it and never says Liftoff!. What's wrong? Circle one.



a. A. There is no base case to stop it
b. B. It should call countdown ((n) + (1))
c. C. Recursion isn't allowed in Scratch

5. Explain your thinking

Explain recursion to a younger student, using an example from real life (like Russian dolls or mirrors). Include the words base case.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can write a recursive procedure with a base case.

☐ I can draw a fractal with recursion.

☐ I can explain why the input must shrink.